

Axiomatic Constraint Compilation and Discrete Mechanics

A Linear Synthesis of Scalar Algebras, Action Emergence, and State-Space Addressability

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Scientific models have historically operated on substance-based ontologies, treating matter, space, and time as foundational entities governed by external physical laws. An alternative paradigm—the constraint-compilation framework—reconstructs physical and computational reality from an axiomatic foundation based on the operational limits of state transitions. By establishing a small set of primitive constraints governing system updates, higher-order physical structures—including local conservation laws, the complex scalar field, Hamilton's principle, gauge symmetries, and discrete permutation mechanics—are derived directly as mathematical necessities.

The core mechanism across all these domains is addressability in change: a constrained system expands its effective capability not by violating its underlying rules, but by discovering new representation spaces and orthogonal channels that exploit latent structural addressability. This report provides a linear synthesis of the complete theoretical framework and empirical corpus. It details the axiomatic pipeline, extracts all compiled theorems and novel quantitative measurements, and outlines the structural audit trail governing the framework's internal consistency.

Primitive Constraint Architecture and Topological State Space

The framework begins with five irreducible primitive constraints that define the admissibility of state space configurations. These constraints do not act as external rules applied to pre-existing objects; rather, they define the topological boundaries outside of which no physical or computational state can exist.

Constraint 0 establishes that distinction exists ($s_A \neq s_B$). It defines the undifferentiated ground state, where total global connection behaves locally as a void. Graph-theoretically, Constraint 0 is represented by a complete graph K_N whose Laplacian matrix $L = D - A$ yields a spectrum of vibrational modes consisting of a single zero eigenvalue with multiplicity 1 (representing global monolithic translation) and an excited

eigenvalue N with multiplicity $N - 1$ (representing maximal connectivity and stiffness). Traversing a boundary walk on a 12-cell cylindrical manifold yields a stasis period of 12 steps across a single face, whereas introducing an offset seam transforms the topology into a Möbius strip with a face-parity twist, forcing a 24-step traversal that serializes superposition into temporal sequence.

Constraint 1 mandates continuation: every admissible state $s \in S$ must admit at least one legal successor $s' \in S$ ($s' \neq s$) under the transition algebra. Terminal states and static fixed points are strictly excluded. Topologically, Constraint 1 mandates a nowhere-vanishing vector field on the compact state manifold M . By the Poincaré-Hopf theorem, the sum of the indices of the vector field's zeros equals the Euler characteristic $\chi(M)$; because zeros are forbidden, the manifold must satisfy $\chi(M) = 0$.

Constraint 1a establishes independence preservation: independent distinctions remain independently continuable. For independent continuation operators T_A and T_B , joint evolution must factor as $T_{AB} = T_A \otimes T_B$, enforcing the algebraic commutation condition $[T_A, T_B] = T_A T_B - T_B T_A = 0$. This single commutator serves as a structural spine operating across three distinct layers: at the transition algebra layer it eliminates coordination channels to enforce local conservation; at the scalar algebra layer it forces commutativity to select the complex numbers; and at the symmetry layer it constrains gauge sector factorization.

Constraint 2 enforces history preservation: every continuation preserves the set of states reachable from prior history, forbidding the deletion of accessible nodes from the transition graph. Iterated transition functions over finite sets yield descending chains $S \supseteq f(S) \supseteq f^2(S) \dots$ which inevitably terminate at a fixed point $f^k(S) = f^{k+1}(S)$. To prevent the flow of possibility from freezing, Constraint 2 forces the physical state space to possess infinite degrees of freedom.

Constraint 5 defines the observable event primitive: an observable event is the minimal continuation simultaneously carrying two independent degrees of freedom, continuous spatial propagation and a distinct temporal residue. Constraint 5 forces the real dimension of the scalar continuation algebra to satisfy $\dim_{\mathbb{R}}(D) \geq 2$, while spontaneous symmetry breaking eliminates the static vacuum as a forbidden fixed point.

Rigorously auditing the primitive set demonstrates that initial formulations contained redundant axioms. At the binary base ($n = 2$), Constraint 1 alone forces the unique legal state transition map to be bijective (the swap map), rendering Constraint 2 derivable rather than primitive at the foundational layer. Similarly, uniform in-degree and isotropic change are derived directly from Constraint 1 when combined with a discrete difference operator.

The Division Algebra Theorem and Complex Continuation Amplitudes

Standard quantum field theory postulates complex-valued probability amplitudes. The Division Algebra Theorem replaces axiomatic postulation with a proof demonstrating that the complex field $\mathbb{C}$ is the unique admissible scalar continuation algebra under the primitive constraint set $\{C_0, C_1, C_{1a}, C_2, C_5\}$.

The mathematical derivation reduces the candidate algebras through a four-stage elimination process:

First, the scalar weight algebra D over the state space must be a finite-dimensional division algebra over $\mathbb{R}$. Constraints 0 and 1 forbid zero divisors, as $d \cdot \psi = 0$ for $d \neq 0, \psi \neq 0$ terminates a lawful state. By

Frobenius' Theorem (1877), the only associative finite-dimensional division algebras over $\mathbb{R}$ are the real numbers $\mathbb{R}$ (dimension 1), the complex numbers $\mathbb{C}$ (dimension 2), and the quaternions $\mathbb{H}$ (dimension 4).

Second, algebras containing zero divisors—such as the direct product algebra $\mathbb{R}^2$ —are physically inadmissible. In $\mathbb{R}^2$, element-wise multiplication yields $(1,0) \times (0,1) = (0,0)$. Two non-zero, lawful continuation amplitudes compose to generate an absolute zero state, creating a Constraint 1-forbidden terminal fixed point.

Third, the real numbers $\mathbb{R}$ are eliminated because $\mathbb{R}$ carries a single real degree of freedom. Its only intrinsic linear transformation generates $Z_2 = \{+1, -1\}$, a disconnected discrete group. Constraint 5 requires continuous spatial phase rotations via $U(1) = \{e^{i\theta} : \theta \in [0, 2\pi)\}$, which cannot be realized over $\mathbb{R}$ without introducing a second real dimension, thereby forcing the transition from $\mathbb{R}$ to $\mathbb{C}$.

Fourth, the quaternions $\mathbb{H}$ are eliminated via algebraic simplicity. $\mathbb{H}$ is non-commutative ($ij = k = -ji$) and simple as a real algebra, possessing only trivial two-sided ideals ($\{0\}$ and $\mathbb{H}$). For any algebra homomorphism $\rho: \mathbb{H} \rightarrow \text{End}(V)$, the kernel $\ker(\rho)$ must be either $\{0\}$ (faithful) or $\mathbb{H}$ (trivial). If faithful, ρ is injective and inherits $\mathbb{H}$'s non-commutativity, yielding $[\rho(i), \rho(j)] = 2\rho(k) \neq 0$. Applying independent phase updates to independent subsystems produces ordering-dependent outcomes, directly violating Constraint 1a. If trivial, ρ maps all elements to zero, exiting $\mathbb{H}$ entirely. No intermediate representation exists.

The complex field $\mathbb{C}$ uniquely survives all compiled constraints, possessing continuous phase rotation $U(1)$, commutativity under Constraint 1a, and exact two-dimensional minimality under Methodological Principle M_1 . $U(1)$ emerges directly as the automorphism group of the compiled scalar field.

Dual-Channel State Spaces, the Wall Theorem, and Structural Channel Taxes

A core conceptual refinement repairs the historical misinterpretation of Constraint 1's prohibition on fixed points. Stasis in a single variable does not constitute a forbidden fixed point provided the complete multi-channel state continues to evolve.

A physical standing wave $u(x, t) = \sin(kx)\cos(\omega t)$ holds a spatial node at x_0 where $\sin(kx_0) = 0$. Empirical measurement confirms that while the node's value channel is pinned at zero with absolute precision, its relational channel (the spatial gradient $\partial_x u$) sweeps a continuous range. The value channel is frozen while the relation channel remains active, satisfying Constraint 1.

Evaluating product states (V, R) across a value channel of size $V = 5$ and a relation channel of size $R = 4$ (20 total states) proves that freezing one channel corresponds to a valid physical measurement fact, whereas freezing both channels collapses the state to a forbidden self-loop:

Configuration / Freeze State	Alive States	Dead States (Self-Loops)	Architectural Status
Unconstrained (Free Field)	20	0	Admissible

Configuration / Freeze State	Alive States	Dead States (Self-Loops)	Architectural Status
Value Channel Frozen Only	20	0	Admissible Fact
Relation Channel Frozen Only	20	0	Admissible Fact
Both Channels Frozen	0	20	FORBIDDEN Fixed Point

The Wall Theorem establishes that a distinction remains admissible under Constraint 1 if and only if at least one of its channels remains unfrozen. The unfrozen channel constitutes the remaining degree of freedom required for physical continuation. Applying Constraint 1 to a symbol stream over a base- k alphabet prohibits adjacent symbol repetitions. Preventing repetition requires inserting exactly one zero-content pointer symbol, which occupies a fixed structural fraction of the channel given by $1/(k + 1)$. Empirical execution across varied bases confirms this structural tax:

Base (k)	Measured Carrier Occupancy	Theoretical $1/(k+1)$	Absolute Error
3	0.249955	0.250000	4.5×10^{-5} [cite: 5]
7	0.125123	0.125000	1.2×10^{-4} [cite: 5]
10	0.091070	0.090909 (1/11)	1.6×10^{-4} [cite: 5]

For a grid of dimension N , spatial parity determines the existence of static reference anchors. Because $N^2 \pmod{4} \in \{0,1\}$, even-dimension grids ($N^2 = 4k$) are fully paired with zero self-paired centers. Odd-dimension grids ($N^2 = 4k + 1$) leave exactly one unpaired center cell mapping to itself under reflection. This center cell acts as a static position anchor (position variance 0.00e + 00) while its substrate continues to oscillate (substrate variance 0.4969 across 19 zero-crossings). Removing this parity anchor causes spatial orbits to drift by 0.5844 over 300 steps.

Discrete Action Emergence and Variational Field Equations

Classical and quantum mechanics postulate Hamilton's principle ($\delta S = 0$). Within this framework, action is compiled directly as the accumulated transition cost of discrete paths under Constraints 0 through 5.

A discrete path $\gamma = (s_0, s_1, \dots, s_n)$ has an accumulated path cost $\mathcal{C}(\gamma) = \sum_i c(s_i, s_{i+1})$. The local cost function $c(s_i, s_{i+1})$ is identical to the compiled free-field Lagrangian density $L = (\partial_t \phi)^2 - c^2 (\nabla \phi)^2$ scaled by local volume elements: $c(s_i, s_{i+1}) = L(\phi(s_i), \partial \phi(s_i)) \cdot \Delta t \cdot \Delta V$. In the continuum limit of the redistribution geometry, discrete path cost converges directly to the continuous action functional $S[\gamma] = \int L(\phi, \partial \phi) d^4x$.

Local Constraint 1-consistency requires that the local transition residual $R(x, t) = \square \phi(x, t) = \partial_t^2 \phi - c^2 \nabla^2 \phi$ vanishes everywhere. A non-zero residual indicates uncompensated local content creation, violating local conservation.

The logical progression links local consistency to global extremality bidirectionally: local Constraint 1-consistency at every point is equivalent to the vanishing of the transition residual ($\square \phi(x, t) = 0$), which is equivalent to satisfying the Euler-Lagrange equations, which is in turn equivalent to the action extremality condition $\delta S[\phi] = 0$. Hamilton's principle is thus proved to be the global manifestation of local Constraint 1-consistency.

The Feynman path integral structure compiles naturally from this foundation:

$$\langle \phi_f | \phi_i \rangle = \int_{\text{admissible}} \mathcal{D}[\phi] \exp\left(\frac{iS[\phi]}{\hbar}\right)$$

Enumerate all paths satisfying Constraint 1 continuation. Assign phase weights derived from the compiled scalar field $\mathbb{C}$. As $\hbar \rightarrow 0$, stationary phase isolates the trajectory minimizing transition residuals ($\delta S = 0$).

Transformation Bias, Structural Gravity, and Phase Geometry

The framework addresses what prevents a system from collapsing into a frozen state without causing it to drift coherently. Admissible persistence requires a transformation bias—a continuous gradient in the cost of change.

Evaluating three dynamical regimes over 20,000 execution steps isolates the necessity of a cost gradient:

Regime	System Mechanics	Net Displacement	Position Variance	Mean Step Size	Architectural Status
Frozen	Hard Lock / Zero Gradient	0.00	0.000	0.0000	FORBIDDEN (Constraint 1 Violation)

Regime	System Mechanics	Net Displacement	Position Variance	Mean Step Size	Architectural Status
Formless	Unconstrained Random Walk	74.46	2376.456	0.7889	Unstable (Identity Dissolves)
Biased	Soft Restoring Gradient	0.91	0.989	0.2432	ADMISSIBLE (Bounded Orbit)

Extending the biased regime walk to 50,000 steps demonstrates that the walker never freezes. Its minimum step size remains strictly positive at 0.000007, with exactly 0.0000% zero-length steps, orbiting its attractor perpetually.

The primitive chain mandates that transformation cost cannot be uniform. A single-valued cost landscape generates zero gradient ($\nabla V = 0$), offering no preferred direction for state updates. A system on a flat landscape is forced into either the frozen pole or the formless pole. To satisfy Constraint 1 while preserving identity, the transformation cost must take the form of a continuous gradient ($\nabla V \neq 0$). Gravity is structurally defined as this required cost gradient: a spatial asymmetry that weights motion into a preferred basin without locking any coordinate.

Phase geometry dictates how open structures maintain dynamic range without self-collision. Flat tiling paths terminate against boundaries after 21 tiles. Maintaining constant dynamic range per turn uniquely selects the logarithmic spiral $r = r_0 e^{-b}$, which maintains a constant area-rate ratio of 0.2845 per turn (compared to Archimedean spirals, which vary fifteen-fold from 53.08 to 3.47). Requiring optimal spatial separation to prevent crowding selects the golden angle of 137.5078° , yielding a peak minimum separation of 1.60191 (compared to $135^\circ \rightarrow 0.34478$ and $144^\circ \rightarrow 0.11172$).

Discrete Mechanics, De Bruijn Repairs, and Permutation Surgeries

Transition dynamics operating on discrete shift registers exhibit exact algebraic properties when mapped to state space cycles.

Linear maximum-length feedback shift registers generating m -sequences over $GF(2)$ fail Constraint 1 continuation due to a unique structural defect: the all-zero state (0^n) maps to itself, forming an absorbing fixed point. Exhaustive search proves that the minimal repair required to convert an m -sequence into a single continuous de Bruijn cycle of length 2^n is a two-state swap between 0^n and its immediate feedback successor 10^{n-1} :

Lane Width (n)	Unpatched State Trajectory	Minimal Patch Swap Coordinates	Resulting Structure	Density Balance (H)
$n = 4$	0^4 loop 15-cycle	$(0000) \leftrightarrow (1000)$	de Bruijn 16-cycle	≈ 0.35 [cite: 6]
$n = 5$	0^5 loop 31-cycle	$(00000) \leftrightarrow (10000)$	de Bruijn 32-cycle	≈ 0.35 [cite: 6]
$n = 6$	0^6 loop 63-cycle	$(000000) \leftrightarrow (100000)$	de Bruijn 64-cycle	≈ 0.35 [cite: 6]
$n = 7$	0^7 loop 127-cycle	$(0000000) \leftrightarrow (1000000)$	de Bruijn 128-cycle	≈ 0.35 [cite: 6]

Evaluating transitions on isolated vertices introduces artificial non-determinism, as vertex projections ($s_0 \rightarrow s_1 \rightarrow s_2$) discard path history, erasing exactly 1 bit of information per path merge. Lifting the state space to edges (s_i, s_{i+1}) recovers strict bijectivity. Organizing state lifts into a discrete jet tower defines higher-order observables: J^0 (Position), J^1 (Edge/Velocity), and J^2 (Triple/Acceleration). Measuring the discrete second difference Δ^2 on a 32-node graph confirms that while acceleration is multi-valued when evaluated on vertices (32 ambiguous sets), it is strictly single-valued when evaluated on edges (0 ambiguous sets across 64 edges).

When composite systems interact across boundaries, the interface generates two distinct accounting metrics governed by non-overlapping algebras:

The Admission Price (A) operates as a static filter on the state space, restricting input to admissible configurations $Adm = \{x \in S: P(x) \neq \text{reject}\}$. The admission cost is $A(I) = -\log_2(|Adm|/|S|)$. Admission is strictly idempotent ($A(I \circ I) = A(I)$) and sub-additive. Sequential validation yields $P(F_2|F_1) = 1.0000$, charging a joint cost of 5.0710 bits compared to a naive linear sum of 10.1419 bits.

The Crossing Price (X) operates as a dynamic meter per execution, governed by conditional entropy $X(I) = H(X|Y) = \sum p(y)H(X|Y = y)$. Crossing cost is strictly additive ($X(g \circ f) = X(f) + X(g)$) across pushforward measure chains.

For linear channel transformations represented by matrix M , physical power efficiency equals $|\det M|$. The erasure cost per pass is $-\log_2|\det M|$ bits. In continuous systems governed by generator matrix K , splitting K into symmetric and skew-symmetric parts proves that skew-symmetric rotational commutators burn zero bits ($\det e^{Kt} = 1.0$). Bit loss rate is driven exclusively by the trace of the symmetric generator: $dB/dt =$

$-\text{tr}(M)/\ln 2$. Zero trace preserves volume (Liouville's theorem), while negative trace contracts phase space volume, driving Landauer information erasure.

Macroscopic state updates are driven by minimal-support 3-cycle permutation surgeries $\tau \in A_\Omega$. State observables decouple into two distinct structural projections: Settlement Mass ($O(\lambda) = \log_2 \text{ord}(\lambda)$), reading global algebraic order, and Transparency ($F(\lambda) = \#\{i: L_i = 1\}$), counting fixed points.

Fixed points are Lipschitz bounded ($||\text{Fix}(\sigma P)| - |\text{Fix}(P)|| \leq |\text{supp } \sigma| \leq 3$), meaning a 3-cycle surgery alters F by at most 3. Conversely, settlement mass O possesses no modulus of continuity; a transposition can split an N -cycle into coprime lengths $s = N/2 - 1$ and $N - s$, exploding the order to $\text{lcm}(s, N - s) \approx N^2/4$, causing relative order jumps scaling as $O(\tau P)/O(P) \sim N/4$.

Permutation dynamics under 3-cycle surgeries generate a deterministic macroscopic drift field $D(p) = E[G] - E[\Lambda]$, balancing prime factor gains (G) from cycle merges against exponent losses (Λ) from cycle splits:

Prime Count (p)	Sample Size (n)	Expected Gain E[G]	Expected Loss E[Λ]	Net Drift D(p)
5	91	11.15	6.46	+5.58 [cite: 6]
6	220	11.38	9.67	+2.52 [cite: 6]
7	394	12.09	10.73	+2.10 [cite: 6]
8	483	11.50	12.08	+0.18 (Equilibrium)
9	423	11.16	12.81	-0.92 [cite: 6]
10	235	9.75	14.19	-3.71 [cite: 6]

Prime Count (p)	Sample Size (n)	Expected Gain E[G]	Expected Loss E[Λ]	Net Drift D(p)
11	98	8.37	15.21	−6.02 [cite: 6]

Expected loss scales linearly with existing prime count (~ 1.3 bits per prime), while gain remains flat. The zero-crossing equilibrium $p^*(W)$ scales with register lane width W as $p^*(W) \approx 0.94W - 2.8$.

Discrete mechanics divides operations into three order-dependent algebraic layers: Constraints (MAY), Execution (DOES), and Observation (SEEN). Restricting then observing (SEEN $\circ$ MAY) disagrees with observing then restricting (MAY $\circ$ SEEN) in 49.1% of trials. This non-commutation of the pipeline constitutes the formal origin of the Entropy Seam—entropy S_π is the structural obstruction to measurement commuting with coarse-graining.

Program Audit Ledger and Failure Taxonomy

Applying Constraint 2 (History Preservation) reflexively to the research program mandates publishing all failed derivations, null results, and retractions. Retaining failure branches prevents the program from merging distinct epistemic states.

The seventeen documented corrections fall into three primary failure modes: nine Overclaims of Generality (stating a narrow true result in a wider false form), four instances of Measuring the Wrong Layer (computing observables on representations rather than underlying relations), and four Misread Nulls (interpreting a flat measurement as absence of effect when the effect moved channels).

The consolidated correction and retraction register documents the program's self-falsification history:

Correction ID	Initial Stated Claim	Measured Empirical Truth	Corrected Architectural Status
R1 [cite: 7]	Feed-forward map $x \mapsto x + \text{core}(x)$ is bijective.	98 unique outputs from 256 inputs at $n = 8$.	RETRACTED. Addition is group-invertible; irreversibility lies strictly in chaining value discard.
R2 [cite: 7]	Prime-indexed constants break cycle closure in SHA-256.	Composite, sequential, random, and zero constants	RETRACTED. Constant primality provides

Correction ID	Initial Stated Claim	Measured Empirical Truth	Corrected Architectural Status
		all yield ideal avalanche (0.5000).	auditability; geometry carries diffusion.
R3 [cite: 7]	Lattices admit no additive valuation unless graded.	Modular valuation space dim 5; meet-homomorphisms into $\mathbb{R}$ dim 0.	RETRACTED. Replaced by idempotence lemma $(f(x \wedge x) = 2f(x) \Rightarrow f = 0)$.
R5 [cite: 7]	Constraint 2 is an independent primitive axiom.	At $n = 2$, exactly 1 Constraint 1-legal map exists, and it is bijective.	RETRACTED. Reversibility is free at $n = 2$; Constraint 2 emerges upon coarse-graining.
R6 [cite: 7]	Constraint 3 (uniform in-degree) is an independent primitive.	Derivable from Constraint 1 combined with discrete difference operator Δ .	RETRACTED. Retained as efficiency condition, not primitive.
N3 [cite: 7]	Floating-point channel separation holds over decimal scaling.	Significand drifts $0.508 \rightarrow 0.951$ under base-10 scaling; exact under base-2.	NARROWED. Channel independence is exact exclusively under radix scaling.
N5 [cite: 7]	Difference-cascade voids are specific to π .	π zero count yields $z = -1.56$ against random digit stream baseline.	NULL. Voids are generic artifacts of the difference operator.
N6 [cite: 7]	$\pi/9$ is a preferred attractor angle in square spirals.	Chi-square = 4.0 in 45° bins; shell restarts cluster at corners ($45^\circ, 135^\circ, 225^\circ, 315^\circ$).	NULL. Four-fold structure is an artifact of the square lattice.

Correction ID	Initial Stated Claim	Measured Empirical Truth	Corrected Architectural Status
N7 [cite: 7]	Aspect ratio is visible in hash digest outputs.	Vertical neighbour agreement = 0.4985 to 0.5019 across nine aspects.	NULL. Mixing consumes grid geometry; redirected focus to trajectories.
A1 [cite: 7, 8]	Floating-point exponent renormalization occurs on ~ 100% of operations.	Measured at 38.7% (38,691/100,000 multiplications).	CORRECTED. Renormalization occurs on main execution path (~ 2/5 ops), not error path.

Synthesis

The compilation chain established across these theoretical and empirical results demonstrates that physical laws represent the compiled requirements of continuation under immutable constraints. State space initialization begins with Constraint 0 ground state cancellations combined with Constraint 1 continuation mandates, forcing admissible state space manifolds to satisfy $\chi(M) = 0$. Algebraic selection via Constraints 0, 1, 1a, 2, and 5 systematically eliminates non-field, one-dimensional, and non-commutative algebras, isolating the complex numbers $\mathbb{C}$ as the unique scalar field for physical amplitudes.

Transition costs on spatial adjacency graphs converge to the free-field Lagrangian, where local Constraint 1-consistency requires a vanishing transition residual ($\phi = 0$), establishing Hamilton's principle ($\delta S = 0$) as a compiled theorem. Maintaining Constraint 1 continuation while preserving identity requires a spatial cost gradient ($\nabla V \neq 0$), identifying gravity as the required transformation bias holding systems away from frozen fixed points. In discrete state spaces, shift register execution forces minimal two-state origin repairs to form de Bruijn cycles, while non-commutation between constraint, execution, and observation layers generates thermodynamic entropy. By tracking state space transitions through explicit algebraic operations, the constraint-compilation framework provides a unified, fully deterministic model of physical and computational mechanics.